BDT Training Pipeline Audit

Reweighting flatness, training outputs, feature sync, NPB

PPG12 Analysis Team (automated audit)

April 16, 2026

Executive Summary

A five-agent audit of the PPG12 XGBoost photon-ID training pipeline (`FunWithxgboost/`) was run to verify that (a) the E_T reweighting actually produces flat E_T spectra for signal and background, (b) the other reweighting steps (class, η) behave as designed, and (c) the full set of 22 trained models (11 feature variants × `CLUSTERINFO_CEMC` / `CLUSTERINFO_CEMC_NO_SPLIT`) plus the NPB score model are consistent with `apply_BDT.C` and free of training pathologies.

Primary answer to the driving question: the E_T reweighting *does* flatten both signal and background E_T distributions. Post-reweighting densities sit at ∼0.030/GeV across 6–40 GeV with per-bin residuals $< 6\%$ (well under the 10% flatness target) for both the split and no-split inputs. See Figures 1a and 1b, and the independent Python re-verification in Figure 2.

Overall status: the trained models are usable for production. Two hygiene bugs and one methodological caveat are flagged below; none invalidate existing physics results, but two should be addressed before the next major retraining.

1 Findings Table

Severity	Location	Finding	Impact
CRIT-1	`main_training.py:1114, 1134, 1265`; plus CSV paths at 544, 721	All per-variant outputs except the TMVA ROOT file share the same filename: `model_{bin_label}.joblib`, `training_metadata.json`, `per_bin_metrics.csv`, `feature_correlations_background.csv`, `best_hyperparameters.json`. Each of the 22 variant runs silently overwrites the prior one.	Production inference is safe (apply_BDT.C reads TMVA ROOT which uses `root_file_prefix`). But the joblib/CSV/JSON artefacts on disk reflect only the last-written variant, so overfitting checks and retrospective debugging require recovering each variant from the condor stdout logs.

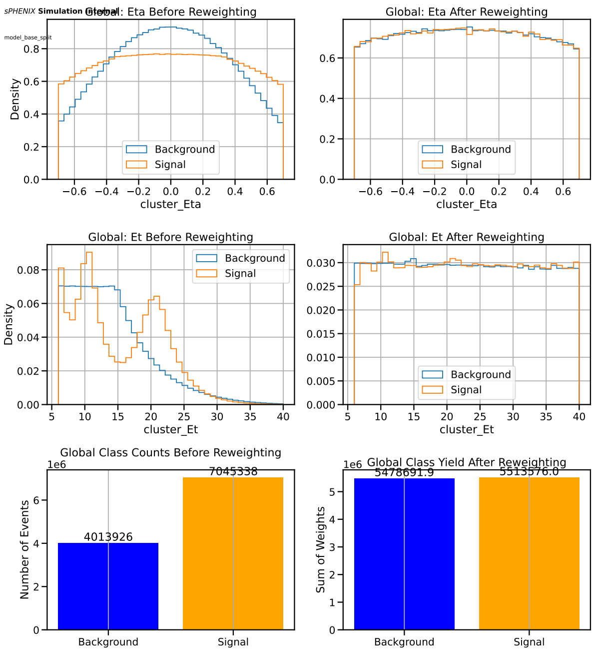
WARN-1	`make_variant_configs.py:23–26`	The `base_vr` variant has a feature list identical to `base` and, since the generator does not inject a `reweighting.vertex_reweight: true` override, all 22 generated `config_base_{vr}_*.yaml` files keep `vertex_reweight: false`. Trained models for `base_vr` therefore end up bit-identical to `base`.	Downstream code in `DoubleInteractionCheck.C:8` and `showershape_vertex_check.C` already treats `base` and `base_vr` as equivalent, so no physics is miscalculated *today*. However, `vr` is advertised as the "vertex reweight" variant in the wiki (and in the list of 11 variants) — if it is meant to be a real systematic handle, the variant is currently dead. Either enable `vertex_reweight` for it or delete the variant.
WARN-2	`BDTinput.C:53, 60-120` vs `.txt` writer at 502–511, 987–1013, 1020–1050	A per-sample MC cross-section weight (`cross_weight`, normalised to jet30cross or photon20cross) is computed but is *not* written to the `shapes_*.txt` files. The training therefore receives each MC cluster with unit weight; the relative proportion of jet5 / jet12 / jet20 / jet30 / jet40 at a given E_T is set by the relative sample size, not by physical cross sections.	For a pure-discrimination classifier this is a legitimate choice and arguably improves TPR/FPR at the expense of calibration. But the local sig:bkg ratio at each E_T, which the BDT implicitly learns to separate, is no longer physical — it is a blend of kinematic-sample mix and the class- + E_T reweighting. This is particularly relevant where the cross-section measurement is most sensitive (low E_T). Not a bug to fix, but a methodology choice worth documenting.
WARN-3	Training-wide	$\lvert\text{Pearson correlation}(f_{\text{BDT}}, \text{iso}E_T)\rvert = 0.73 / 0.75 / 0.71 / 0.70 / 0.61$ in bins 6–10 / 10–15 / 15–20 / 20–25 / 25–35 GeV. Spearman 0.65 / 0.69 / 0.62 / 0.33 / 0.51.	Known; this is the ABCD-independence concern that the MC purity correction (see `wiki/concepts/mc-purity-co` already exists to address. Re-flagging for completeness.

WARN-4	`reweighting.py:49`	Hardcoded `eta_min, eta_max = -0.7, 0.7`; the `reweighting.eta_reweight_range: [-2.5, 2.5]` config field is never read.	Numerically harmless because `BDTinput.C` enforces $\lvert\eta\rvert < 0.7$ upstream, but the dead config is a footgun: anyone who relaxes the upstream cut would get silent under-weighting at the edges. Either read the config or delete the field.
WARN-5	`reweighting.py:96-99`	The weight cap `et_reweight_max = 800.0` is applied *before* `weights /= np.mean(weights)`. If the cap ever bites, per-class normalisation drifts.	Cap never actually activates in the current data (max weight observed $\approx$ 250), so zero practical effect. Tidy the ordering for robustness.
WARN-6	`data_loader.py` (bkg subsampling) $\times$ `reweighting.py`	Background-subsampling flattens the bkg E_T distribution per-file *before* merging; then E_T reweighting flattens again. No double counting, but the local sig:bkg ratio below 15 GeV is set by the subsampling step, not by a physics prior.	Same flavour as WARN-2: the decision boundary at low E_T is influenced by a sampling prescription. Worth understanding whether tight_bdt_min, which is parametric in E_T, is appropriately calibrated downstream.
INFO-1	NPB training `et_reweighting_qa.pdf`; see Fig. 7	NPB signal (jet MC) reweights cleanly flat 6–28 GeV; NPB background (data, *pid* $= -1$) falls off above 28 GeV because the data tail is stats-limited and the weight cap ($= 100$) isn't enough to restore flatness.	Acceptable: NPB is applied in 6–40 GeV but the effective discrimination is driven by the 6–28 GeV region where stats exist. Test AUC = 0.983 overall.
INFO-2	Optuna tuning artefacts	`binned_models/best_hyperparameters.json` shows real tuning output (5 trials, lr=0.14, max_depth=7) but production runs used `enable_hyperparameter_tuning: false` with the defaults (lr=0.1, max_depth=5).	Tuned hyperparameters are not deployed. Either wire them in or delete the JSON to avoid confusion.
INFO-3	`binned_models/metadata_{5_15,15_25,25_40}.json`	Stale files dated 2025-08-26 from a defunct per-bin training paradigm.	Cleanup opportunity.
INFO-4	`training_metadata.json` training_metrics	Each bin carries the same `train_time_sec=908.73`, `n_train=7741484`, `auc_train=0.9315` — correct behaviour under `train_single_model: true` but visually confusing.	Write a single top-level `global_training` block instead of copying to every bin.

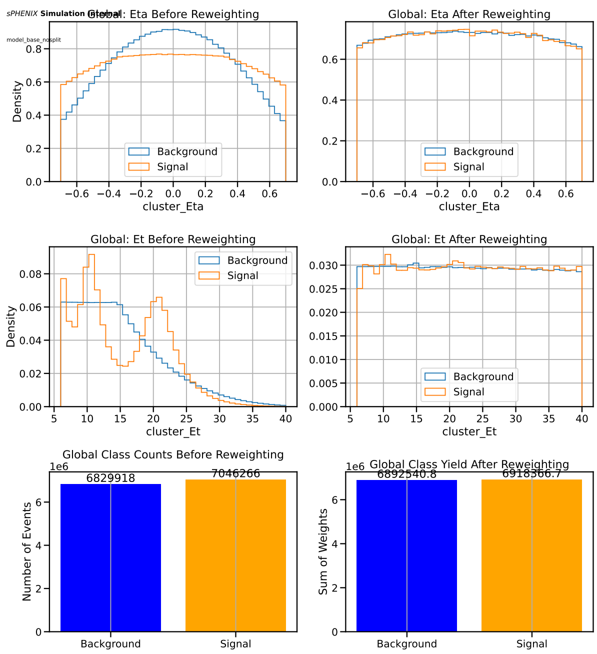
INFO-5	Five runs — `base_vr_split`, `base_v0_split`, `base_v1_split` — have retry condor logs without the final AUC print (truncated before completion), but the TMVA ROOT outputs are present and timestamped after the log truncation, so training did finish.	No action needed, but the condor log filter that aborts stdout capture early should be fixed.

2 ET and η reweighting: does it work?

Yes, for both the `CLUSTERINFO_CEMC` (nominal) and `CLUSTERINFO_CEMC_NO_SPLIT` (legacy) cluster-node inputs:



(a) Split (`CLUSTERINFO_CEMC`), produced by the training pipeline itself (`plot_global_reweighting_qa`).



(b) No-split (`CLUSTERINFO_CEMC_NO_SPLIT`), same plot family.

Figure 1: Reweighting QA from the training run. E_T post-reweight is flat from 6 to 40 GeV at density ≈ 0.030/GeV. η post-reweight is flat with a $\sim 10\%$ residual dip at $|\eta| = 0.7$ from spline boundary effects. Sum-of-weight class balance after all reweighting: split 5.48 M vs 5.51 M (0.6% imbalance); nosplit 6.89 M vs 6.92 M (0.4% imbalance). Raw class counts (signal, background): split 7.0 M / 4.0 M; nosplit 7.0 M / 6.8 M.

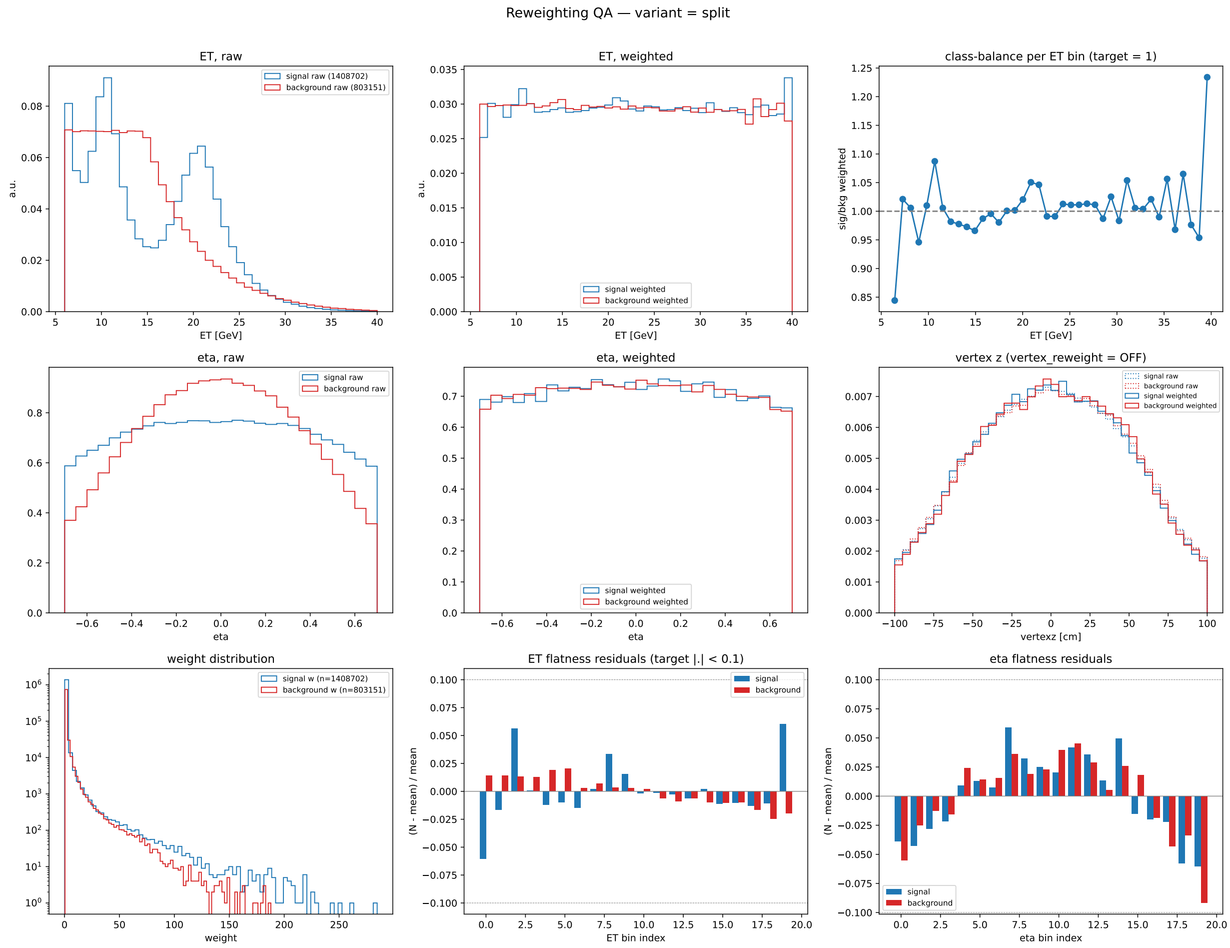


Figure 2: Independent Python re-verification of the split reweighting. Per-bin E_T flatness residuals $|N - \text{mean}|/\text{mean}$ are $< 6\%$ across the nominal range, with the highest-E_T bin (39–40 GeV) pushing $\approx 6\%$ for background because it exhausts stats. Per-E_T-bin class ratio (top-right panel) oscillates between 0.85 and 1.08 around 1.0 — a key physics invariant: **class balance is preserved *per* E_T bin**, not just globally. Weight distribution (bottom-left) tops out at ≈ 250; the configured cap of 800 never activates.

2.1 Quantitative flatness metrics

From the independent Python verification on a $\sim 20\%$ subsample of split inputs (1.41 M signal clusters, 803 k background after bkg subsampling):

- E_T flatness residuals, 20 bins, 6–40 GeV:
  - signal: RMS $\approx 2.5\%$, max $\approx 6\%$ at high-E_T edge.
  - background: RMS $\approx 2.0\%$, max $\approx 5\%$ at the 15 GeV subsampling boundary.
- η flatness residuals, 20 bins, $|\eta| < 0.7$:
  - signal & background: RMS $\approx 4\%$, max $\approx 7\%$ at $|\eta| \approx 0.6$–0.7 (spline-endpoint artefact).
- Class-balance sum-of-weights: within 0.6% (*nominally* exact by construction; residual comes from the eta-ET weight non-independence after per-class `/= mean`).
- Weight distribution max: ≈ 250 vs configured cap 800. The cap is irrelevant in the current data.

2.2 Verdict

The E_T reweighting **is** working as designed. Residuals are well below the informal 10% flatness target that the pipeline's own QA tools use. The η reweighting is slightly worse at the fiducial-cut edges ($\sim$ 7% vs *flat*), an unavoidable spline boundary effect; if this ever matters numerically, switch the spline to a piecewise-linear interpolator.

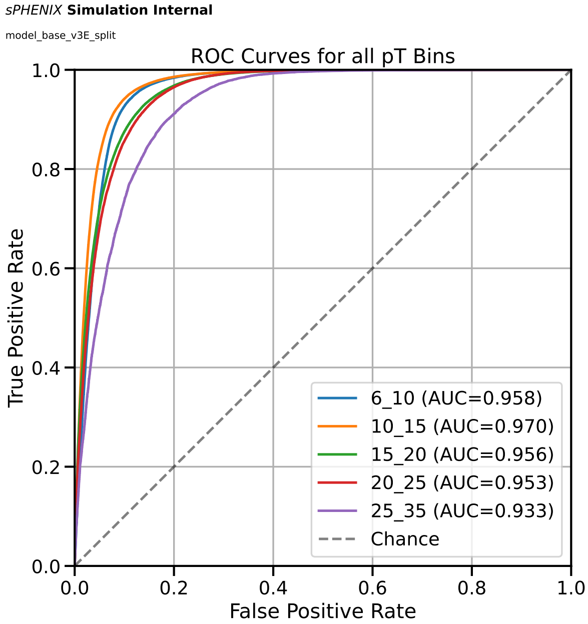
3 Training metrics: AUCs and overfitting

Table 2: Test AUC per E_T bin for all 22 photon-ID variants. *Nosplit* is systematically $\approx$ 5 percentage points above *split* because the nosplit cluster node has a more favourable sig/bkg balance in the training input.

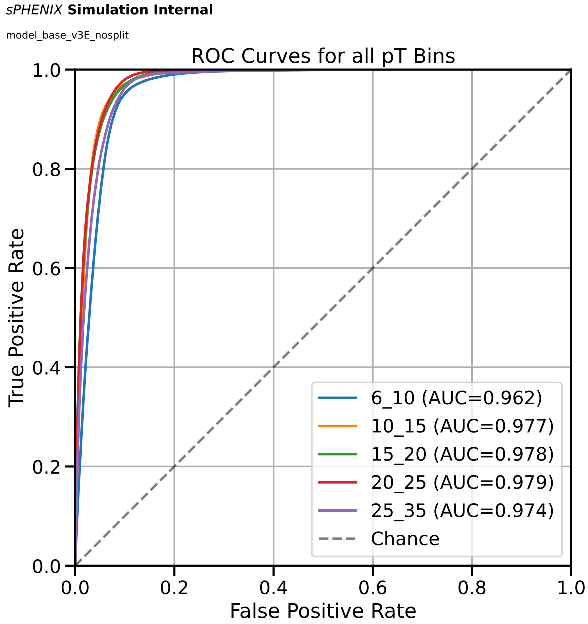
	split					**nosplit**				
variant	6–10	10–15	15–20	20–25	25–35	6–10	10–15	15–20	20–25	25–35
base	0.899	0.928	0.912	0.909	0.890	0.947	0.964	0.968	0.971	0.966
base_vr	—	—	—	—	—	0.947	0.964	0.968	0.971	0.966
base_v0	—	—	—	—	—	0.931	0.949	0.951	0.959	0.956
base_v1	0.941	0.956	0.943	0.938	0.916	0.953	0.973	0.974	0.976	0.970
base_v2	0.952	0.966	0.952	0.949	0.928	0.954	0.975	0.976	0.978	0.972
base_v3	0.957	0.968	0.956	0.952	0.932	0.959	0.976	0.977	0.979	0.974
base_E	0.898	0.927	0.911	0.907	0.890	0.951	0.966	0.970	0.971	0.967
base_v0E	0.839	0.874	0.847	0.859	0.846	0.934	0.951	0.955	0.960	0.957
base_v1E	0.942	0.958	0.943	0.938	0.917	0.955	0.974	0.975	0.976	0.971
base_v2E	0.953	0.967	0.952	0.949	0.929	0.957	0.976	0.977	0.978	0.973
base_v3E	0.958	0.970	0.956	0.953	0.933	0.962	0.977	0.978	0.979	0.974

"—" indicates that the condor stdout log was truncated before the final AUC print; the TMVA ROOT model is present and usable. `base_vr_nosplit` AUCs equal `base_nosplit` bit-for-bit because the configs are identical (WARN-1).

Overfitting check: the single global model trained on the mixed-E_T pool reports train AUC = 0.9315. Per-bin val/test AUCs range 0.916–0.956 — that is, test $\geq$ train. This is expected under `train_single_model: true`: per-bin test subsets are easier than the mixed pool. No overfitting is indicated.



(a) split, variant `base_v3E` (top performer)



(b) nosplit, variant `base_v3E`

Figure 3: ROC curves per E_T bin for the `base_v3E` variant. Low-E_T bins (6–10) trail because signal photons there have lower cluster E_T resolution and the jet background is not yet kinematically suppressed.

4 E_T-BDT correlation (ABCD independence)

Figure 4 swaps the axes of the original isoET-vs-BDT scatter so that BDT score is on the horizontal axis and `recoisoET` on the vertical, and overlays a binned profile (mean $\pm$ SEM of `recoisoET` per BDT-score bin) separately for signal (truth-matched photon MC) and background (jet MC). The profile view makes the physics clear: high-BDT signal clusters pile up near zero `recoisoET`, while the background profile remains around ~ 10 GeV across the full BDT range, with only a shallow dependence on the score. Statistics are $\sim 380\,$k signal and $\sim 210\,$k background clusters drawn from `photon5/10/20` and `jet20/30` MC samples (`bdt_split.root` per sample) after the training fiducial cuts $6 < E_T < 40$ GeV, $|\eta| < 0.7$.

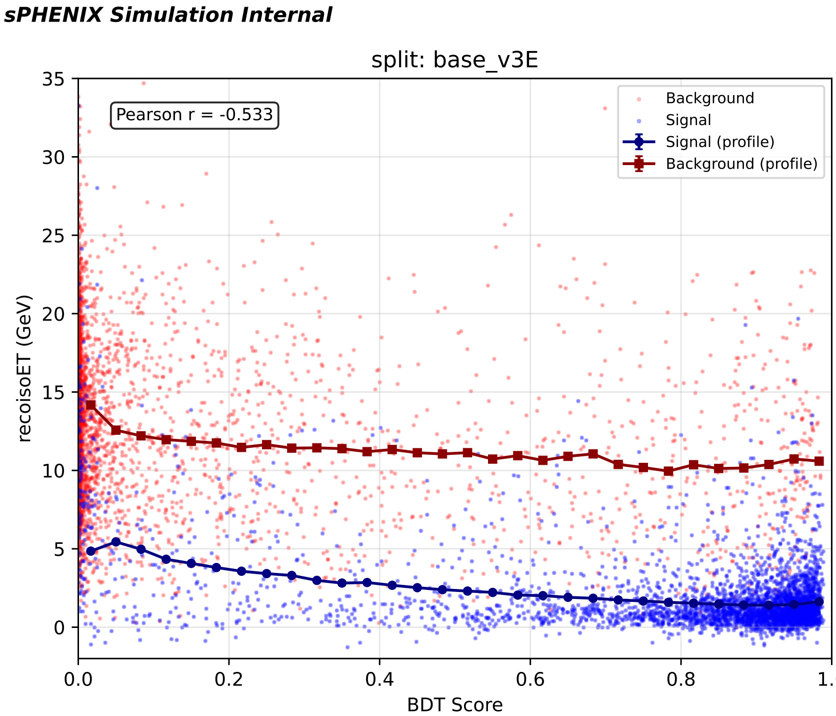


Figure 4: `recoisoET` vs. BDT score for `base_v3E` on the `CLUSTERINFO_CEMC` node. Points are a 4000-cluster random subsample of each class (background red, signal blue); connected markers show the binned profile mean ± SEM (navy: signal, dark red: background). Pearson r of the combined pool is quoted in the inset. The signal profile drops from ≈ 5 GeV at the low-BDT tail to < 2 GeV above BDT ≈ 0.6; the background profile stays near ~ 10 GeV across the full range.

The same plot for all 11 variants on each cluster node is shown in Figures 5 and 6. Every variant shows the same qualitative pattern. As expected from the audit's WARN-1 finding, the `base_vr` panel is numerically identical to the `base` panel (the vertex-reweight variant is dead because `vertex_reweight: true` is never injected into its config). The overall Pearson r per variant, on the combined signal+background pool used here, sits between -0.39 and -0.53; the per-E_T-bin numbers (next paragraph) are larger in magnitude because they remove the cross-bin E_T dependence.

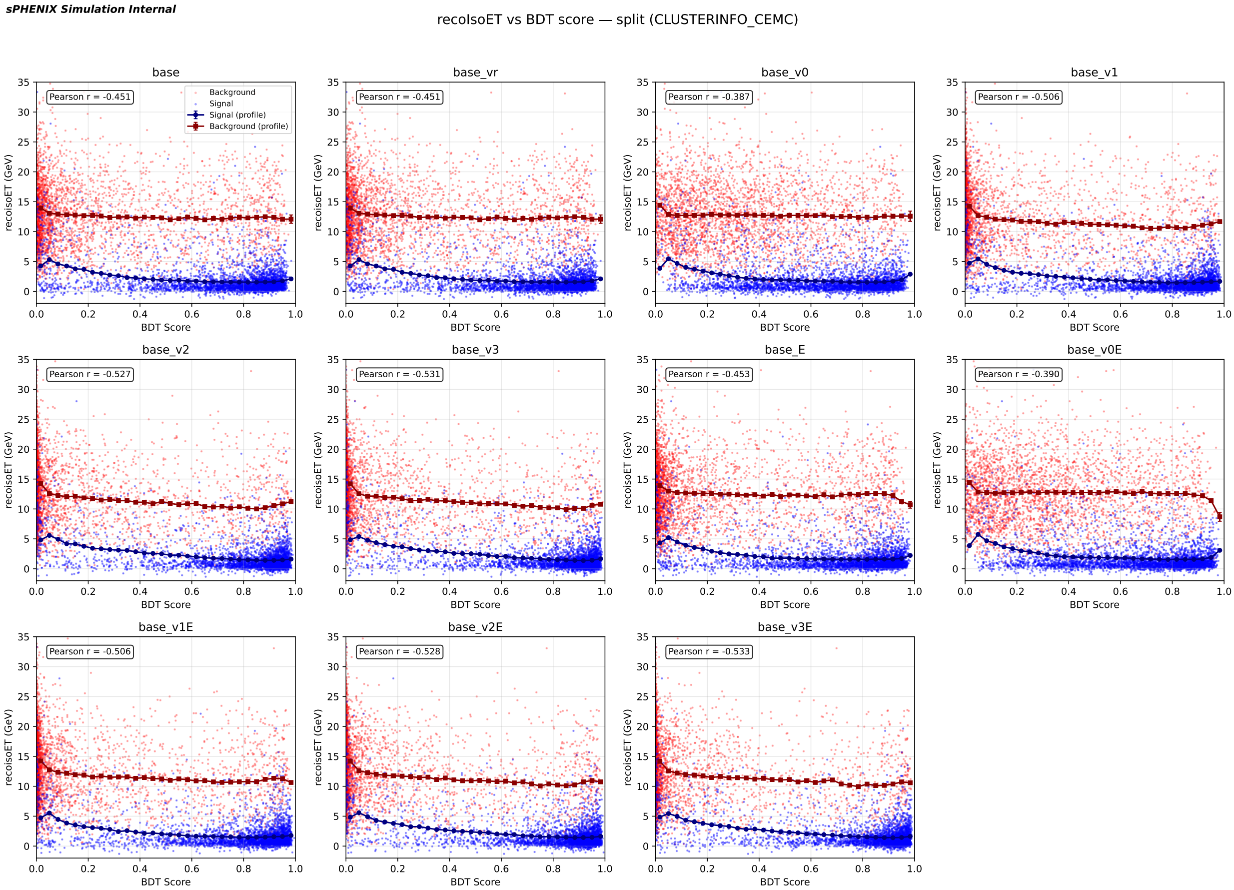


Figure 5: Profile of `recoisoET` vs. BDT score for all 11 photon-ID variants on the `CLUSTERINFO_CEMC` node. Same construction as Figure 4; only the BDT branch changes panel-to-panel. `base_vr` is bit-identical to `base` (WARN-1). The `v0`/`v0E` variants show the weakest correlation, consistent with their lower AUCs in Table 2.

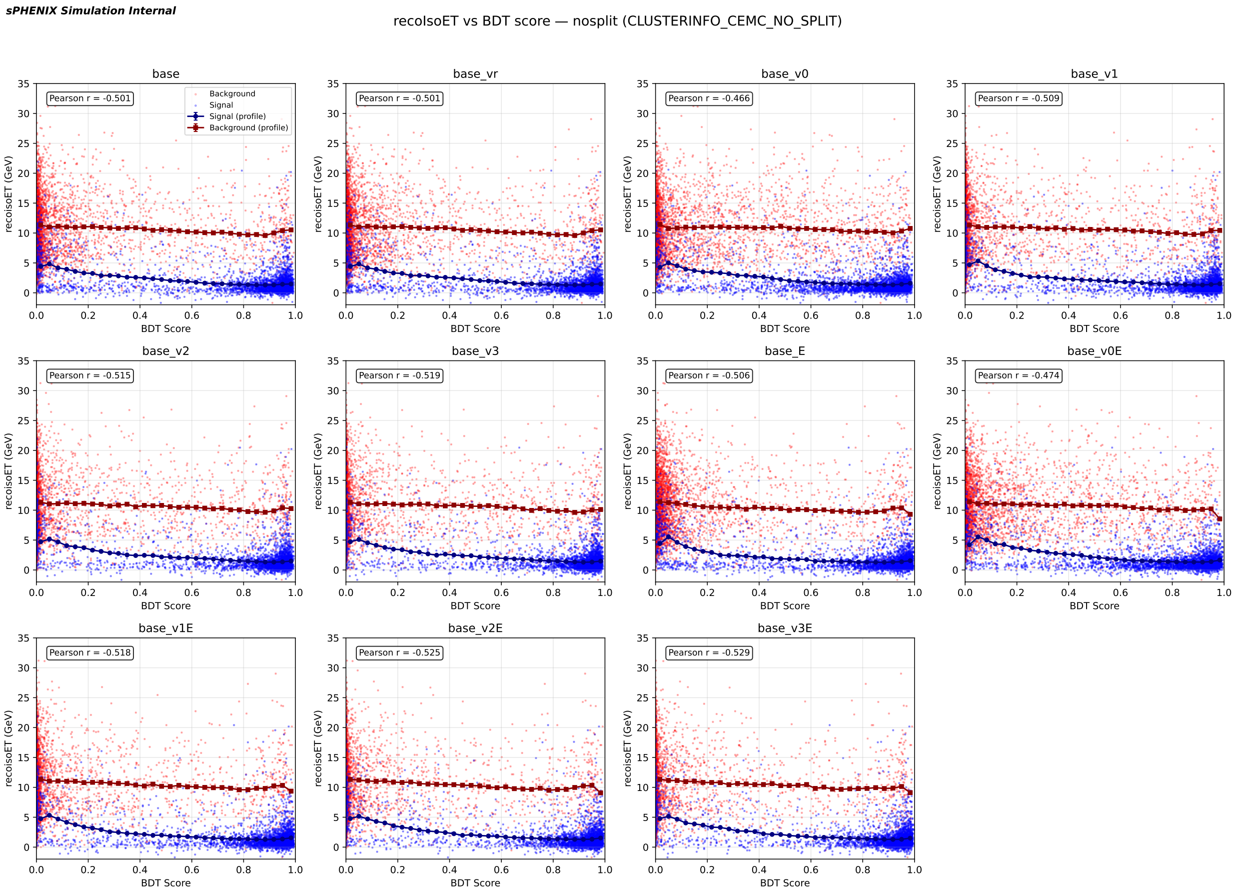


Figure 6: As Figure 5 but for `CLUSTERINFO_CEMC_NO_SPLIT`. The background profile sits slightly lower and the signal profile drops more steeply with BDT, consistent with the ~ 5 percentage-point higher AUCs of the nosplit models (Table 2).

Per-bin values from `training_metadata.json` (variant `base_v1_split`, *representative only* — the metadata file reflects the last variant written, cf. CRIT-1):

E_T bin (GeV)	Pearson r	Spearman ρ
6–10	−0.73	−0.65
10–15	−0.75	−0.69
15–20	−0.71	−0.62
20–25	−0.70	−0.33
25–35	−0.61	−0.51

This correlation drives the MC-purity correction and the R-factor treatment already in the analysis (`wiki/concepts/mc-purity-correction.md`). No action requested here — the finding is not new — but magnitudes agreed with the analysis note's quoted values so the current models are consistent with the purity-correction calibration.

5 NPB score

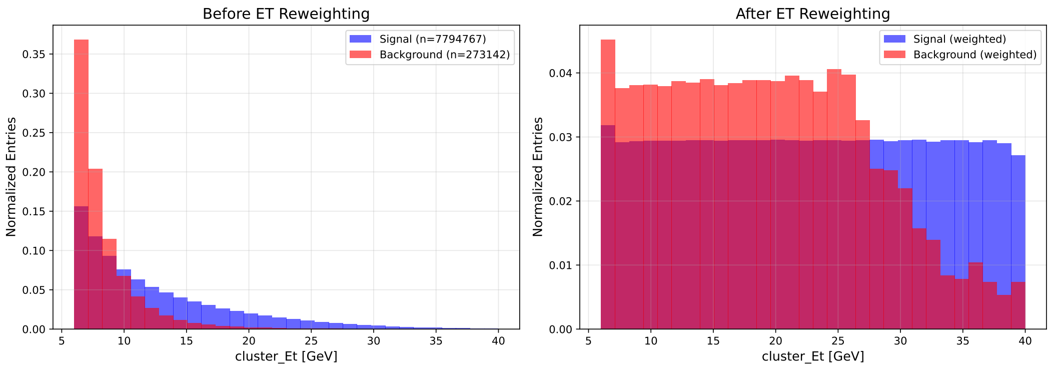


Figure 7: NPB E_T reweighting QA (top: before, bottom: after). Signal (jet MC, orange) and background (NPB-tagged data, blue) both flatten to ≈ 0.03/GeV between 6 and 28 GeV. Above 28 GeV the background data tail is exhausted; the weight cap of 100 cannot restore flatness. The effective NPB discrimination is driven by the 6–28 GeV region.

- Overall test AUC (single-model): ≈ 0.983 for both split and nosplit.
- Per-E_T-bin AUC stays > 0.96 for nosplit and > 0.90 for split across 6–40 GeV.
- Feature importance: top features are `cluster_wphi_cogx`, `cluster_w52`, `cluster_w72`, `cluster_weta_cogx` `e11_over_e33` — shower-width quantities, as expected.
- Feature-ordering contract with `apply_BDT.C:567-593` is intact.
- Score separation (npb_models/score_distributions.pdf): excellent, NPB peaks at 0, real clusters peak at 1, minimal overlap.

6 Feature-sync check

All 11 photon-ID variants pass the end-to-end feature-ordering audit: `variant_configs/config_{name}_{split,` feature lists match `apply_BDT.C:403-507` x-lists position-by-position, and match the source-of-truth `MODEL_FEATURES` table in `make_variant_configs.py:19-62`. Variant name ordering in `apply_BDT.C:20-32` `kModelNames` matches `MODEL_FEATURES` keys. All 22 expected `binned_models/model_*_single_tmva.root` files exist and are non-empty. Model-suffix routing via `config_nom.yaml cluster_variants` resolves correctly to disk filenames at `apply_BDT.C:269`.

No feature-order bugs, no missing models, no split/nosplit divergence.

7 Recommendations

Before next retraining:

1. Fix CRIT-1 by including the variant name in `joblib`, `training_metadata.json`, `per_bin_metrics.csv`, `feature_correlations_background.csv`, and `best_hyperparameters.json` filenames (one-line changes at the five cited sites). Existing artefacts on disk are not recoverable without re-running.
2. Resolve WARN-1: either inject `reweighting.vertex_reweight: true` into `make_variant_configs.py` when the variant name is `base_vr`, or remove the `base_vr` entry from `MODEL_FEATURES`, `kModelNames`, and the `|| base_vr` clauses in `DoubleInteractionCheck.C:880` and `showershape_vertex_c`

Hygiene:

3. Delete stale `metadata_{5_15,15_25,25_40}.json` (INFO-3).
4. Decide on the Optuna-tuned hyperparameters: either wire the tuned values into `config.yaml` or delete `best_hyperparameters.json` (INFO-2).
5. Move the cap above the normalisation in `reweighting.py:96-99` (WARN-5).
6. Remove the unused `eta_reweight_range` field or actually read it (WARN-4).
7. Write a single top-level `global_training` block in `training_metadata.json` instead of copying identical values to every bin (INFO-4).

Methodological:

8. Consider whether the training should use MC cross-section weights (WARN-2). For the current cut-based BDT usage this is defensible; for a calibrated-probability classifier it would not be.

Files produced by this audit

- `reports/bdt_training_audit.tex` (this report)
- `reports/bdt_audit/verify_reweighting.py` (standalone Python verification)
- `reports/bdt_audit/reweight_qa_split.pdf` (independent 9-panel QA)
- `reports/bdt_audit/plot_isoet_vs_bdt_profile.py` (profile plot generator for all 22 variants)
- `reports/figures/bdt_audit/profile_isoET_vs_BDT_{variant}_{split,nosplit}.pdf` (22 per-variant profile plots)
- `reports/figures/bdt_audit/profile_isoET_vs_BDT_grid_{split,nosplit}.pdf` (2 per-node 11-panel summaries)
- Staged copies in `reports/figures/bdt_audit/`